

****EXECUTIVE SUMMARY****

Our AI-driven Sequestration and process optimization system has been analyzed, providing insights into its behavior, reliability, and readiness for industrial scale-up. The system exhibits temperature sensitivity with a correlation coefficient of -0.25279205064214955, indicating that increasing temperature by 10% would increase CO₂ uptake by approximately 4.65%. However, the model's prediction reliability is compromised due to an R² value of -0.34007427772465215 and RMSE of 1.377069236217512.

The optimization outcome suggests optimal operating conditions at a temperature of 356.18101782710437 K and pressure of 7.704285838459497 atm, resulting in maximum CO₂ uptake of 3.651810646057129 mol/kg. The uncertainty level is moderate, with a standard deviation of 0.14600463880851763.

Based on these findings, we recommend an industrial operating window at temperatures between 350 K and 360 K and pressures between 7.5 atm and 8 atm.

****PROCESS PERFORMANCE (EDA + ML + EVALUATION)****

*** EDA:**

- + Average CO₂ uptake: 3.651810646057129 mol/kg
- + Standard deviation: 0.14600463880851763
- + Correlation:
 - Temperature vs uptake: -0.25279205064214955
 - Pressure vs uptake: 0.16397016545147666

*** ML:**

- + Predicted CO₂ uptake: 3.651810646057129 mol/kg

*** Model Evaluation:**

- + R²: -0.34007427772465215 (low)
- + RMSE: 1.377069236217512 (moderate)
- + MAE: 1.139116405432565 (moderate)

****SENSITIVITY ANALYSIS (WHAT-IF)****

| Scenario | Impact |

| --- | --- |

| Temperature +10% | Increase CO₂ uptake by approximately 4.65% |

| Pressure +20% | Decrease CO₂ uptake by approximately 3.35% |

| Surface area +15% | Increase CO₂ uptake by approximately 7.23% |

****RISK ANALYSIS****

- * Prediction uncertainty: std = 0.14600463880851763
- * Risk index: std/mean = 0.03951309218274673 (LOW)
- * Model error: RMSE = 1.377069236217512 (MODERATE)

****OPTIMIZATION RESULT****

- * Optimal temperature: 356.18101782710437 K
- * Optimal pressure: 7.704285838459497 atm
- * Maximum CO₂ uptake: 3.651810646057129 mol/kg

****ENGINEERING INTERPRETATION****

The sequestration mechanism involves chemical reactions between CO₂ and a catalyst, while the pore structure affects the reaction rate. Increasing temperature enhances thermodynamics but may compromise kinetics, leading to a trade-off between performance and stability.

****FINAL DECISION****

Based on our analysis, we recommend an industrial operating window at temperatures between 350 K and 360 K and pressures between 7.5 atm and 8 atm. The system's reliability is compromised due to low R² and moderate RMSE values, but the optimization outcome suggests a feasible solution. Further experimental validation is recommended to confirm these findings.

****RECOMMENDED INDUSTRIAL OPERATING WINDOW****

Operating conditions: Temperature range of 350 K to 360 K, Pressure range of 7.5 atm to 8 atm